
Assignment: REST API & JavaScript Fetch

Objective

Build a small application that fetches data from an API and displays it on the screen.

Research Questions (Find Answers Yourself)

Part A: API Fundamentals

1. What is an API?
2. Why do frontend applications need APIs?
3. What is REST API?
4. What is the difference between:
 - GET
 - POST
 - PUT
 - DELETE
5. Why are URLs usually written as:

```
/users  
/products  
/orders
```

instead of:

```
/getUsers  
/getProducts
```

6. What is the meaning of:

- 200
- 201
- 400
- 401
- 404
- 500

Part B: JavaScript

1. What is Promise?
2. What is async/await?
3. What happens if an API call fails?
4. Difference between:

`.then()`

And

`async/await`

5. What is JSON?
6. Why do we use:

`JSON.parse()`
`JSON.stringify()`

Practical Assignment

Use: <https://jsonplaceholder.typicode.com/users>

Task 1: Fetch Users

Create a page that shows:

- Name
- Email
- Phone

Requirements:

- Use Fetch API
 - Show data in table format
-

Task 2: Loading State

Before data loads show:

Loading...

Question:

Why is loading state required?

Task 3: Error Handling

If API fails show:

Something went wrong

Questions:

- What error did you receive?
 - How did you handle it?
-

Task 4: Search User

Add search box.

Requirements:

- Search by Name
- Search by Email

Questions:

- Which JavaScript method did you use?
 - Why?
-

Task 5: Create User

Create a form:

Name
Email
Phone

Add user to UI.

Questions:

1. Why is POST used?
 2. Why does JSONPlaceholder not permanently save data?
-

Deliverables

- [GitHub Repository](#)
- [Screenshots](#)
- [Answers Document](#)



Assignment: Browser Storage

Objective

Understand how browsers store data.

Research Questions

1. What is LocalStorage?
 2. What is SessionStorage?
 3. What is Cookie?
 4. Which one survives browser restart?
 5. Which one is sent to server automatically?
 6. What is HttpOnly?
 7. What is XSS attack?
 8. Why is LocalStorage vulnerable to XSS?
-

JavaScript Tasks

Task 1: Save User Preference

Create:

Theme Selector

Options:

- Dark
- Light

Save selected theme.

Questions

1. Which storage did you use?

2. Why?
 3. What happens after page refresh?
-

Task 2: Recently Visited Pages

Store:

Home
Users
Profile

in LocalStorage.

Display visit history.

JavaScript Concepts to Explore

Research:

```
localStorage.setItem()  
localStorage.getItem()  
localStorage.removeItem()  
localStorage.clear()
```

Task 3: Counter Application

Create:

Increment Button
Decrement Button

Store count in LocalStorage.

After refresh:

Count should remain.

Task 4: Shopping Cart Simulation

Store products in LocalStorage.

Requirements:

- Add Product
- Remove Product
- Show Cart Count

Research:

```
JSON.stringify()  
JSON.parse()
```

Why are they required?

Assignment: Authentication & JWT

Objective

Understand how login systems work.

Research Questions

1. What is Authentication?
 2. What is Authorization?
 3. What is Session?
 4. What is JWT?
 5. Why do modern SPAs use JWT?
 6. What are the three parts of JWT?
 7. Why should passwords never be stored in JWT?
 8. What is Refresh Token?
 9. What is Access Token?
-

Practical Assignment

Task 1: JWT Investigation

Visit:

<https://jwt.io>

Find:

- Header
 - Payload
 - Signature
-

Questions

1. Which algorithm is used?
2. Can payload be read without secret key?

3. Is JWT encrypted?
-

Task 2: Login Page

Create:

Email

Password

Login Button

Validation:

- Email required
 - Password required
-

Task 3: Protected Dashboard

After login:

Show Dashboard.

Without token:

Redirect to Login.

Research Questions

1. How does protected routing work?
 2. How do websites know a user is logged in?
-

Task 4: Logout

Requirements:

- Remove token

- Redirect Login

Questions:

1. Why remove token?
 2. What security issue occurs if token remains?
-

Challenge (Important)

Build a Mini User Management System

Features:

- ✓ Login Page
- ✓ Store JWT Token
- ✓ Protected Dashboard
- ✓ Fetch Users from API
- ✓ Search Users
- ✓ Persist Theme Using LocalStorage
- ✓ Logout

Concepts Should be Covered

- HTML
- CSS
- JavaScript
- Fetch API
- Async/Await
- Error Handling
- LocalStorage
- JWT
- Authentication
- Protected Routes